

Kun Zhang

PhD Applicant — Large Language Models: Post-Training, Alignment, and Evaluation

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My work is on large language models, across evaluation validity, mechanistic interpretability, and alignment. What holds it together is the experimental design rather than the topic. Before I report a positive result I build the control that would produce it for the wrong reason. **Several of the papers below end in a claimed effect that does not survive that test.**

Education

University of Leeds

Leeds, United Kingdom

MSc Advanced Computer Science — taught and assessed entirely in English

Sep 2025 – Sep 2026

- Dissertation: a leakage-controlled audit of whether financial text adds forecasting value beyond price and firm-level baselines.
- Selected module marks: Deep Learning **84** • Knowledge Representation & Reasoning **81** • Data Mining & Text Analytics **81** • Machine Learning **77**

Hunan University of Finance and Economics

Changsha, China

BSc Computer Science and Technology

Sep 2021 – Jun 2025

- Final-year project: GAN-based face image inpainting

Publications

All papers below are **first-authored**; five accepted, two under review. Papers marked † were co-authored with my MSc supervisor Dr Chunwei Xia as corresponding author; the remaining two are **sole-authored**. Venue ranks: ICORE2026 (CORE).

ACCEPTED

- [1] K. Zhang, C. Xia†. *Same Exam, Different Scores: A Tokenizer-Grounded Audit of Evaluation-Protocol Sensitivity on Chinese Benchmarks*. **NLPCC 2026** (Resources and Evaluation track) — **Oral**, Springer LNCS.

Pre-registered a mechanistic prediction that first-token-argmax scoring makes the same benchmark assign systematically different scores to the same model. The predicted effect held; the reported ranking is an artefact of the protocol.

- [2] K. Zhang, C. Xia†. *A Point-in-Time Audit of Graph-Snapshot Leakage in GNN Stock Selection*. **PRICAI 2026 CORE B** — regular paper (12–16 pp.), **Oral**, Springer LNAI.

Rebuilt institutional co-holding graphs from raw SEC 13F filings without look-ahead. Most reported gains from graph-snapshot methods do not survive the correction.

- [3] K. Zhang. *PCP: A Controlled Benchmark for Pragmatic Intention Recognition in Clinical Communication*. **LUHME 2026 @ EMNLP** — **Oral**. **Sole author**.

A 3,533-item benchmark for recovering intended meaning from hedging, indirectness and implicature. Eleven systems sit near ceiling on the controlled split (98.7–99.8%) but fall to 79.7–90.6% on externally grounded cases, so a broad controlled probe can hide the difficulty that close distractors expose.

- [4] K. Zhang. *Grounding or Lexical Overlap? A Matched-Pair Study of Internals-Based Context-Faithfulness Probes*. **GroundLM 2026** (workshop at EMNLP) — workshop **Findings** track. **Sole author**.

In the standard extractive setup the faithfulness label correlates with lexical overlap at $r = 0.99$, so a grounding probe can be a string matcher. A matched construction halves the confound ($r = 0.50$); on RAGTruth the probe direction scores below NLI and lexical overlap.

- [5] K. Zhang, C. Xia†. *ProbeShift: Predicting Probe-Direction Rotation from Stability is Largely a Sampling-Noise Floor*. **ACML 2026 CORE C**, conference track — **Oral**, PMLR.

A benchmark and a symmetric-null validity standard for probe-stability signals. ID-only stability predicts a probe's rotation under semantic shift at $\rho = 0.945$, yet scores *higher* on a zero-shift placebo ($\rho = 0.966$); of seven candidate signals only one survives the null.

UNDER REVIEW

Two further first-author manuscripts are under review, both now in Phase 2; titles and venues are withheld during anonymous review.

Research Themes

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- **Evaluation validity and leakage control** [1] [2] [3] — when a reported score measures the protocol, a look-ahead or a controlled split rather than the capability.
 - **Mechanistic interpretability** [4] [5] — causal interventions on model internals, each paired with a control that would reproduce the observation for a different reason.
 - **Alignment and safety** [under review] — which properties a post-training run installs, which it only appears to, and what evidence tells the two apart.

Industry Experience

Chinte Technology

Changsha, China

AI Application Development (intern) — built a retrieval-augmented generation architecture

Nov 2024 – Jun 2025

Guangdong Lianhe Information Technology

China

Data Analysis (intern) — business data evaluation

Feb 2024 – Aug 2024

Academic Service

Reviewer NLPCC 2026 (main conference) • GroundLM 2026 (workshop at EMNLP) • VLM4RWD 2026 (workshop at NeurIPS) 2026

Student member ACL • CCF (China Computer Federation)

Awards

- Apr 2024 **National First Prize**, 2nd “Waiwen Award” National College English Vocabulary Competition — **English Majors Division**. China Foreign Languages Publishing Administration.
Awarded in the English Majors division as a Computer Science student.
- Jul 2024 **Second Prize (national)**, 7th National College Students’ Arts Festival — Ensemble Performance, Group A. **Ministry of Education of the PRC**.
- 2023–24 **Second-Class Scholarship**, Hunan University of Finance and Economics.

Competitions

Kaggle Competitions Expert — 1,957 of 217,591 (top 0.9%) • **Notebooks Expert** — 89 of 60,915 (top 0.1%)

Both medals won **solo**: **silver**, NVIDIA Nemotron Model Reasoning Challenge (**101 of 4,185**, top 2.4%); **bronze**, Orbit Wars (**409 of 4,729**, top 8.6%).

Technical

Frameworks PyTorch, HuggingFace Transformers, scikit-learn, PyTorch Geometric

Referees and their contact details are available on request.